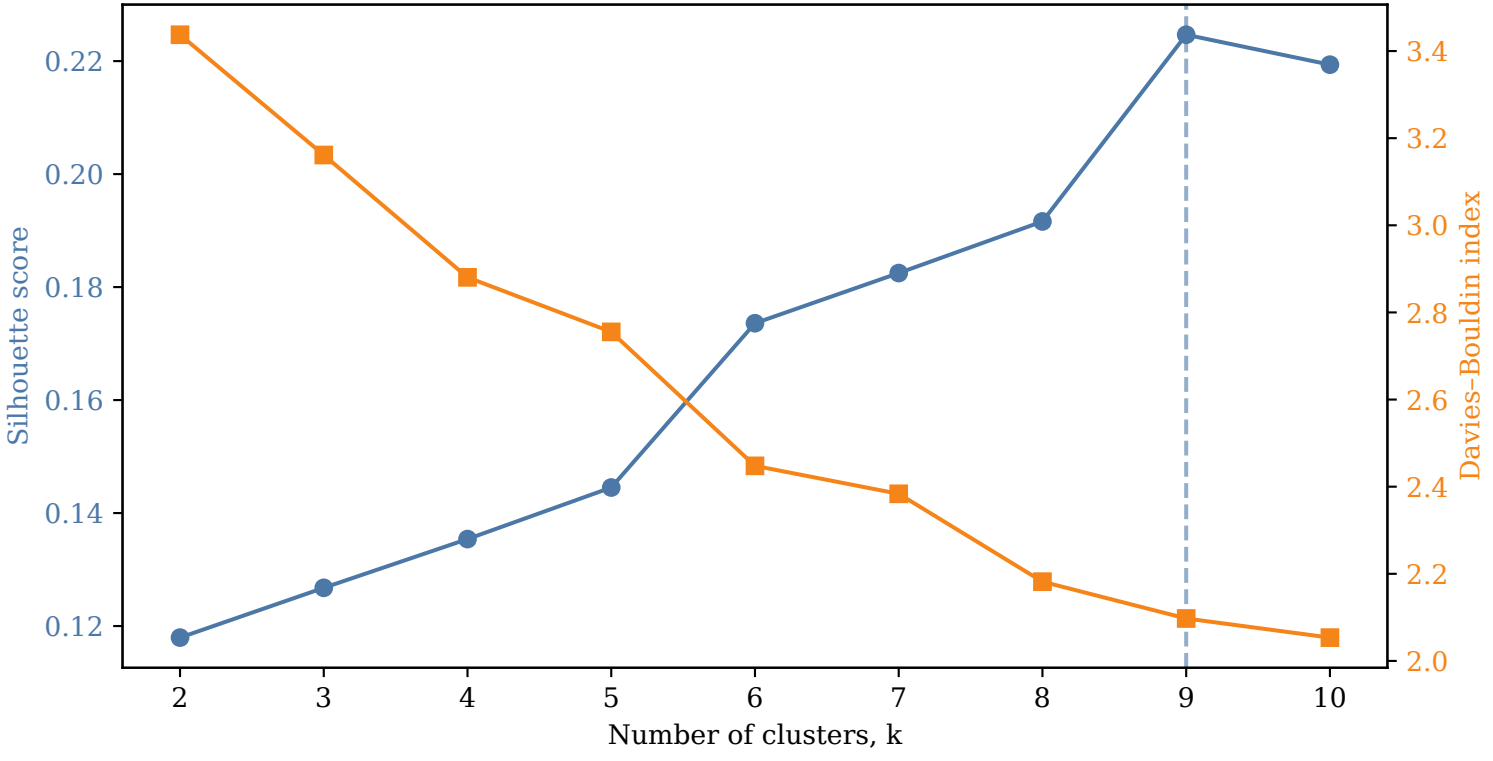

k = 9 maximizes silhouette; the scores indicate exploratory, overlapping groups.